DO NOW — RECALL

Boolean identities and De Morgan's laws

Name: _____ Class: _____ Date: _____

1. Using h(k)=k MOD 5, what is the starting bucket for 17?

2. What does an adjacency list store for a vertex?

3. What is the result of TRUE AND FALSE?

4. Simplify A ∧ 1.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2.

2. Model answer

The neighbouring vertices.

3. Model answer

FALSE.

4. Model answer

A. AND with true leaves A unchanged.

Vocabulary — know these terms

Boolean identity — Rule for simplifying logical expressions.

De Morgan's law — Negate each operand and swap AND and OR.

complement — Logical NOT of a value or expression.

double negation — ¬¬A simplifies to A.

equivalence — Two expressions give the same output for every combination of inputs.

NEW LEARNING

Read, annotate and explain

Negate the inputs

Distribute NOT across the brackets by negating A and B separately.

Swap the operator

When applying De Morgan's law, AND changes to OR; OR changes to AND.

Check equivalence

Both expressions produce the same output for all four combinations of A and B.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Simplify ¬(A ∨ B) and verify its truth table.

Predict and explain

Negate A and B separately and replace OR with AND.

Check and extend

Both expressions are true only when A=0 and B=0, proving equivalence.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Simplify A ∨ 0.

Question 2

Use De Morgan to simplify ¬(P ∧ Q).

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Simplify ¬(A ∨ ¬B) and justify the final result.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

De Morgan's law allows us to remove a NOT from brackets without changing AND or OR.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.